



## Hypothesis Testing *Difference in means test*



## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5$$

## Hypothesis Testing *Difference in means test*

**Claim to be tested...**

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

**Step 1 : Formulate Hypothesis**

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5$$

↑

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5$$

↑



## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5$$

$$H_A: \mu_{\text{Men}} - \mu_{\text{Women}} \neq 12.5$$

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5 \quad \circ \text{ "Difference in means" test}$$

$$H_A: \mu_{\text{Men}} - \mu_{\text{Women}} \neq 12.5$$





## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$\begin{aligned} H_0: \mu_{\text{Men}} - \mu_{\text{Women}} &= 12.5 & \circ \text{ "Difference in means" test} \\ H_A: \mu_{\text{Men}} - \mu_{\text{Women}} &\neq 12.5 & \circ \text{ Two tailed test} \end{aligned}$$

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$\begin{aligned} H_0: \mu_{\text{Men}} - \mu_{\text{Women}} &= 12.5 && \circ \text{ "Difference in means" test} \\ H_A: \mu_{\text{Men}} - \mu_{\text{Women}} &\neq 12.5 && \circ \text{ Two tailed test} \end{aligned}$$

### Step 2 : Calculate the t-statistic

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$\begin{aligned} H_0: \mu_{\text{Men}} - \mu_{\text{Women}} &= 12.5 && \circ \text{ "Difference in means" test} \\ H_A: \mu_{\text{Men}} - \mu_{\text{Women}} &\neq 12.5 && \circ \text{ Two tailed test} \end{aligned}$$

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$\begin{aligned} H_0: \mu_{\text{Men}} - \mu_{\text{Women}} &= 12.5 && \circ \text{ "Difference in means" test} \\ H_A: \mu_{\text{Men}} - \mu_{\text{Women}} &\neq 12.5 && \circ \text{ Two tailed test} \end{aligned}$$

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$\begin{aligned} H_0: \mu_{\text{Men}} - \mu_{\text{Women}} &= 12.5 && \circ \text{"Difference in means" test} \\ H_A: \mu_{\text{Men}} - \mu_{\text{Women}} &\neq 12.5 && \circ \text{Two tailed test} \end{aligned}$$

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations

## Hypothesis Testing *Difference in means test*

### Claim to be tested...

Difference between the population mean height of Men and Women  
Olympic athletes is 12.5 cm

### Step 1 : Formulate Hypothesis

$$H_0: \mu_{\text{Men}} - \mu_{\text{Women}} = 12.5 \quad \circ \text{ "Difference in means" test}$$
$$H_A: \mu_{\text{Men}} - \mu_{\text{Women}} \neq 12.5 \quad \circ \text{ Two tailed test}$$

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.

a thought  
exercise

## Hypothesis Testing *Difference in means test*

### **Step 2 : Calculate the t-statistic**

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- **Subjective judgement**

## Hypothesis Testing *Difference in means test*

### **Step 2 : Calculate the t-statistic**

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*



## Hypothesis Testing *Difference in means test*

### **Step 2 : Calculate the t-statistic**

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\overline{x_1} - \overline{x_2} - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\overline{x_1} - \overline{x_2} - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

#### "Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

#### "Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$



## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

## Hypothesis Testing *Difference in means test*

### Step 2 : Calculate the t-statistic

- Nature of variation or variance in the two populations.
- Is the variance more similar or more dissimilar.
- Need to assume either equal variance or unequal variance across the two populations.
- Subjective judgement ...*assume equal variances*
- In most applications conclusions do not change based on "equal" or "unequal" variance assumption.

"Equal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

"Unequal" variance assumption

$$\frac{\bar{x}_1 - \bar{x}_2 - \mu}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$